

XLS-65

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TokenEscrow

Exit Lane.

An open-ended vault promises withdrawal at any time.
It can no longer keep that promise once it works well.

Open on paper, closed in practice

100%

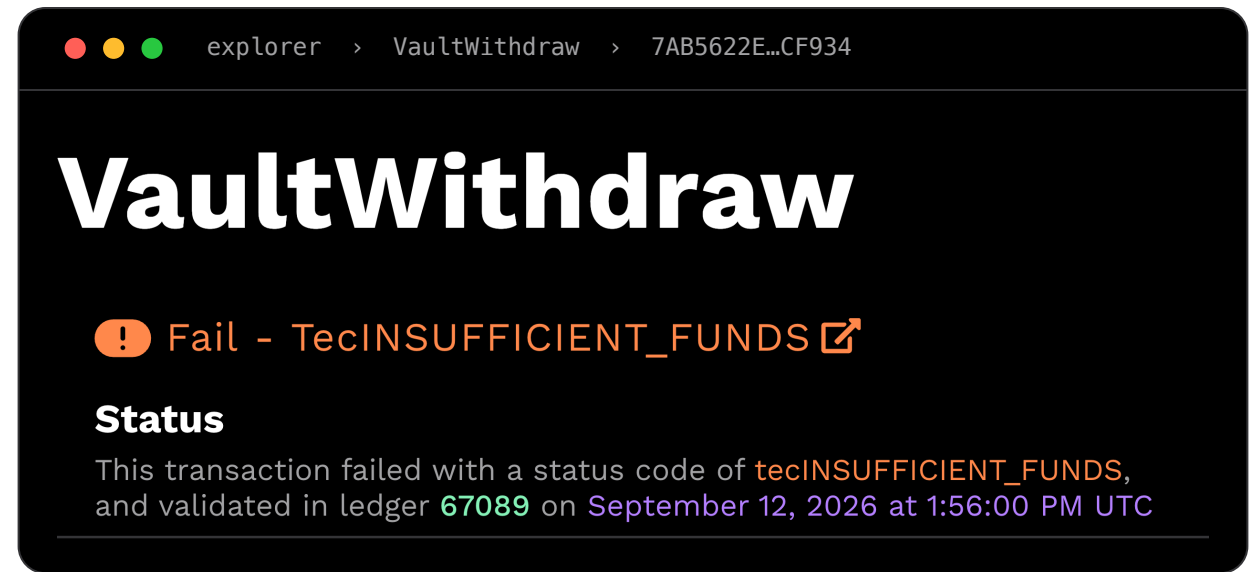
of the vault lent by a single LoanSet

0 XRP

withdrawable by the depositor

4 months

until the last payment is due



The depositor's withdrawal: refused, the loan took the whole vault.

Sell the vault share, not the loan



Price: 97% of share value.

One SHA-256 condition for both escrows.

Live demo.

13 transactions in 56 seconds,
on the devnet, live.

```
~/xrpl-lending $ node scripts/demo.mjs --auto

VaultCreate          tesSUCCESS
VaultDeposit         tesSUCCESS
LoanBrokerSet        tesSUCCESS
LoanBrokerCoverDeposit tesSUCCESS
LoanSet 100%         tesSUCCESS
● VaultWithdraw       tecINSUFFICIENT_FUNDS
MPTokenAuthorize     tesSUCCESS
EscrowCreate x2      tesSUCCESS
EscrowFinish x2      tesSUCCESS
LoanPay              tesSUCCESS
VaultWithdraw        tesSUCCESS

Total time 00:56
```

Minimum bar: 8 of 8

```
README.md > one hash per step

1 Open-ended vault      VaultCreate
2 Deposit                VaultDeposit
3 Broker + cover        LoanBrokerSet
4 Origination + drawdown LoanSet
5 Repayment             LoanPay
6 Principal + yield      VaultWithdraw
7 Guardrails triggered   12 × tec in the README
8 Use case               Exit Lane

share value 1.000000000 → 1.006443880
```

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XLS-65/66 types submitted to the ledger

0

missing types in xrpl@4.6.0

fixCleanup3_4_0: enabled, missing from xrpl.org

Two lending behaviours change on the devnet.

```
LoanManage tfLoanImpair > due date T

• T - 31 s    tecT00_SOON
• T - 13 s    tecT00_SOON
  T +  5 s    tesSUCCESS
```

Impairment refused before the due date, accepted 5 s after.

```
LoanSet > CounterpartySignature

• signLoanSetByCounterparty
  Counterparty: Invalid signature.
  encodeForSigningCounterparty
    tesSUCCESS    23631C36...
```

The xrpl@4.6.0 helper is rejected, the new encoding passes.

Manage a Loan tutorial

xrpl@4.6.0

Proposed fix

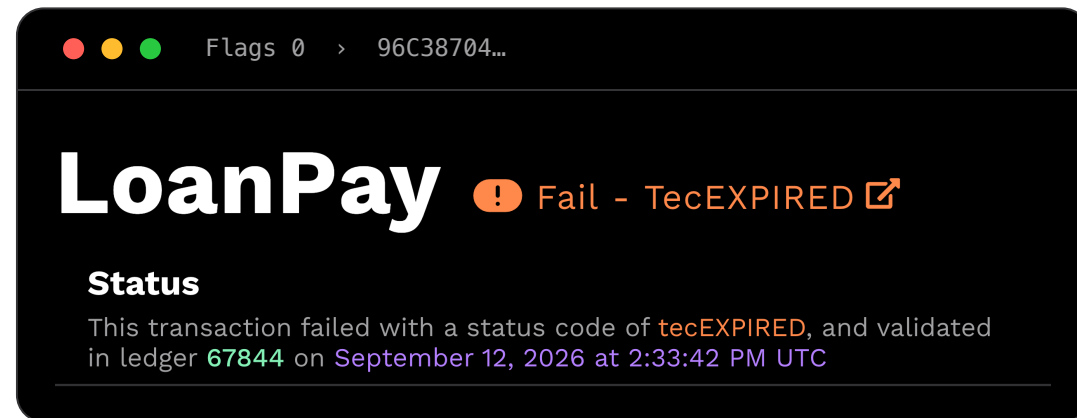
“impair a loan before a payment due date passes”

signLoanSetByCounterparty signs with the previous prefix.

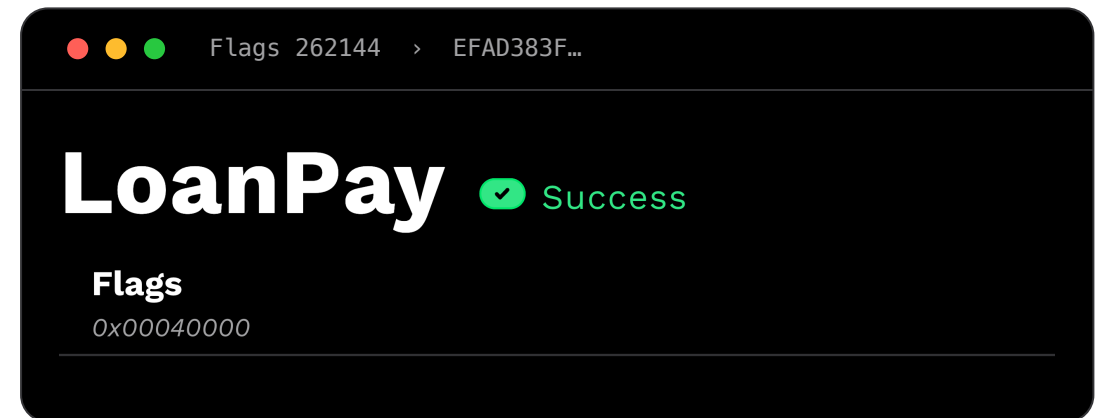
list the amendment, sign with encodeForSigningCounterparty.

A late LoanPay requires tfLoanLatePayment

Same loan, same amount, past its due date.



Same loan, same amount, no flag: tecEXPIRED.



Same loan, same amount, with tfLoanLatePayment: success.

XLS-66, §3.11.4.2

LoanPay reference

Proposed fix

the failure is defined (condition 11).

eight error codes listed on xrpl.org, tecEXPIRED is not one of them.

tecEXPIRED in the LoanPay reference, flag named in the explorer.

First-loss capital: 0.5% of a defaulted loan

1 XRP

of cover posted by the broker

0.02 XRP

taken from the cover at default

3.98 XRP

lost by the depositor, 79.6%

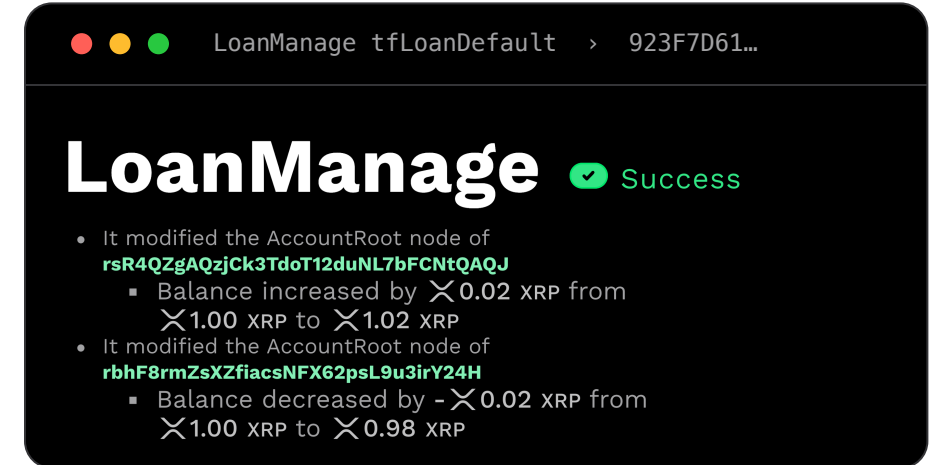
$4 \text{ XRP debt} \times \text{CoverRateMinimum } 10\% \times \text{CoverRateLiquidation } 5\%$

Same ratio on a 50 XRP loan: 0.25 XRP.

Proposed fix

Show the share of a default the cover absorbs.

A delay on LoanBrokerCoverWithdraw after a default.



Default: 0.02 XRP taken from the broker's cover.



21 s later, the broker withdraws the remaining cover.

Six more findings

4 Sole-holder exception missing from xrpl.org

LossUnrealized

6 Four codes, two or three causes each

tecINSUFFICIENT_FUNDS

8 Shares carry IsfMPTCanTrade, OfferCreate refused

temDISABLED

5 No loan_info or loan_broker_info

unknownCmd

7 V1.1 LoanBrokerSet restriction reverted

tesSUCCESS

9 No flag closes a vault to deposits

tecLIMIT_EXCEEDED

Category, severity, repro steps and proposed fix for each in the report.

Nine findings, three pages

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hashes cited, re-checked on the ledger before submission

github.com/charlyppr/xrpl-lending

FEEDBACK.md

FEEDBACK.pdf